FIELD & TRADE REFERENCE · HEALTHCARE

wirelessLAN PROFESSIONALS

Why you cannot design a hospital like an office

A hospital is the one building where a Wi-Fi mistake can contribute to a patient-safety event - life-critical traffic, a protected telemetry band, EMC-regulated devices sharing the air, and a stack of authorities no single WLAN cert teaches.

WMTS: the protected telemetry band most Wi-Fi pros have never heard of

Three bands · 14 MHz total

TV channel 37 · radio astronomy, already quiet	licensed & interference-managed	separate from your Wi-Fi
608-614 MHz	1395-1400 MHz	1427-1432 MHz

lower frequency — higher frequency

schematic spacing, not to frequency scale

A licensed, protected service the FCC created in 2000 so patient telemetry stopped riding vacant TV channels and having to accept interference. It is a separate radio system from the hospital Wi-Fi - not all patient monitoring lives on your network, so never design over it blind.

WHY THE OFFICE PLAYBOOK FAILS

Zero-tolerance roaming

A dropped association or a slow re-auth is a **missed alarm**, not a buffering annoyance. Clinicians, mobile carts, and telemetry must roam **seamlessly facility-wide with no dead zones** - stairwells, elevators, and back-of-house included. Overlapping cells, fast-roaming (802.11r/k/v), voice-grade RF.

RTLS and telemetry multiply AP count

Designs are graded **Data → Voice → Voice/RTLS**, each needing more APs. Voice/RTLS grade (location to ~5 m) demands far higher density than data grade - quote a hospital at data-grade density and it fails. RTLS of assets, staff, and patients often rides the same infrastructure.

Medical-device EMC

Devices are tested to tolerate a defined RF environment under **IEC 60601-1-2** - "tested to a limit" is not "immune to anything." A dense, high-EIRP footprint can push local field strength past what a nearby monitor or pump was qualified for. Coexistence awareness; route EMC to clinical engineering.

THE FOUR-AUTHORITY STACK (RECOGNIZE AND DEFER)

HIPAA · HHS

Data: PHI confidentiality, integrity, and transmission security.

Design for it (encryption, access control, audit logging, segmentation). defer to the privacy / security officer

FDA

Wireless inside the medical device: QoS, coexistence, security, EMC.

Provide a sane RF environment. defer device qualification to the maker + biomed

FCC

Spectrum: WMTS licensing / coordination; unlicensed 2.4/5/6 GHz rules.

Respect WMTS as protected; design Wi-Fi within Part 15. WMTS coordinator

TJC

Accreditation: Environment of Care, alarm-management, ICRA/ILSM in construction.

Coordinate; construction access via the codes & safety reference. the AHJ + facility

THE ONE HANDOFF

Coordinate with clinical / biomedical engineering first. They own the medical devices, their EMC posture, and the telemetry systems - the single most important handoff before you touch a clinical RF environment.

Reference only - recognize and defer. EMC goes to biomed and the device maker, PHI policy to compliance, spectrum coordination to the WMTS coordinator, code and access to the AHJ and facility. The Toolbox asserts no compliance ruling; it teaches you to recognize the environment you are walking into.